

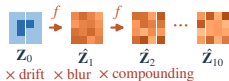
(a) Move, don't regenerate



DINOv2 patches Z_0

+ actions $\rightarrow Z_{1:10}$

Existing WMs regenerate every patch



ShiftWM moves what it has seen



✓ parallel from measured Z_0 ✓ plug-in head

$$\begin{aligned}\hat{Z}_k = & (1-g)Z_0 \text{ keep} \\ & + gT_k \text{ move} \\ & + r_k \text{ correct}\end{aligned}$$

(b) One held-out DROID window

true $t+10$ (decoded)



Direct



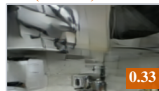
AR-TF (DINO-WM-style)



ShiftWM (ours)



AR (recursive)



Persistence (copy t)



Each tile: that model's own forecast, one shared decoder.
Corner: feature error at $t+10$ (lower is better).

(c) Held-out results

beats Direct in 100/100 DROID horizon-motion bins

